Beyond Feature Attribution: Are Deep Learning Models Learning Hydrological Processes?

Supplementary Information

Contents

1 Experimental Design and Detailed Settings

1.1 Learned Core Circuits Reflect Hydrological Knowledge Across Basins

To analyze how learned core interaction circuits vary under different hydro-climatic conditions, we constructed multiple climate classification indicators based on static basin attributes, including aridity class, snow regime, and geo-climatic type[1]. These classifications were used for post-hoc grouping and statistical analysis of interaction circuits across basins.

Aridity Class

Dryness conditions were classified using the aridity index (`aridity`) through binning[2]. The classification criteria are summarized in Table S1.

Table S1: Aridity-based hydro-climatic classification

Aridity range	**Class**	**Description**
$-1 \leq aridity < 0.65$	Humid	Humid
$0.65 \leq aridity < 1.0$	Sub-humid	Sub-humid
$aridity \geq 1.0$	Arid	Arid

The thresholds (0.65 and 1.0) follow established hydro-climatic stratification practice and effectively distinguish energy-limited from water-limited basins[3]. They provide physically meaningful groupings for subsequent analysis of interaction circuit distributions.

Snow Regime

Snow regime was classified based on snowfall fraction (`frac_snow`) to distinguish rain-dominated, mixed, and snow-influenced basins, as summarized in Table S2[4].

Table S2: Snow regime classification based on snowfall fraction

Fraction range	**Class**	**Description**
$-0.1 \leq frac\_snow < 0.1$	Rain-dominant	Rain-dominated
$0.1 \leq frac\_snow < 0.3$	Mixed	Mixed
$frac\_snow \geq 0.3$	Snow-influenced	Snow-influenced

The selected thresholds (0.1 and 0.3) reflect the gradual transition from rain-driven to snow-driven runoff. Low snowfall fractions correspond to basins dominated by direct rainfall, whereas higher fractions indicate increasing importance of snow accumulation and melt processes[5].

Geo-climatic Type

Geo-climatic type was determined using a hierarchical rule based on elevation (`forcing_elev_m`), longitude (`lon`), and aridity[3]. The pseudocode is shown in Algorithm 1.

This hierarchical classification first identifies high-elevation regions ($\geq 1500m$), then distinguishes oceanic versus inland climates based on longitude, and finally separates arid continental basins using a higher aridity threshold (1.5). This approach provides a physically interpretable stratification of basins across major hydro-climatic regimes.

Algorithm 1 Hierarchical classification for geo-climatic type

Require: Elevation $forcing\_elev\_m$, Longitude lon, Aridity $aridity$
Ensure: Geo-climatic type $geo\_climate\_type$

```
1: if forcing_elev_m ≥ 1500 then
2:     geo_climate_type ← "Plateau/Mountain"
3: else if lon ≤ −120 then
4:     geo_climate_type ← "Oceanic"
5: else if aridity ≥ 1.5 then
6:     geo_climate_type ← "Arid Continental"
7: else
8:     geo_climate_type ← "Humid Continental/Subtropical"
9: end if
```

1.2 Supplementary Notes: Accurate Volume Response but Limited Sensitivity to Temporal and State Perturbations

To examine whether the model encodes physically consistent hydrological sensitivities—rather than relying on spurious statistical correlations—we designed a station-wise counterfactual sensitivity framework on the test split[6]. Let f_θ denote the trained sparse transformer, $\mathbf{X}_i \in \mathbb{R}^{N_i \times L \times F}$ the input tensor for station i (with N_i samples, lookback length L, and F predictors), and $\mathbf{y}_i \in \mathbb{R}^{N_i}$ the observed discharge target. For each intervention setting s, we define

$$\hat{\mathbf{y}}_i^{(0)} = f_\theta(\mathbf{X}_i), \qquad \hat{\mathbf{y}}_i^{(s)} = f_\theta(\mathcal{T}_s(\mathbf{X}_i)), \tag{S1}$$

where $\mathcal{T}_s(\cdot)$ denotes an intervention operator applied in physical space and subsequently mapped back to the normalized model space using the same normalization statistics as in training[7]. This paired design isolates the model's response to controlled perturbations while eliminating cross-sample confounding effects[8].

Experiment design

We considered four intervention families to disentangle model sensitivities to hydrological drivers, including volume, temporal distribution, catchment state, and thermal forcing[9]. The perturbation ranges are chosen to represent moderate yet plausible deviations that remain within the support of the training distribution.

(1) Rainfall scaling. Let P_t denote precipitation at lag $t \in \{1, \ldots, L\}$. With scaling factor α, the intervention is

$$P'_t = \alpha P_t, \qquad \alpha \in \{0.8, 0.9, 1.1, 1.2\}. \tag{S2}$$

This modifies storm magnitude while preserving the original temporal structure of rainfall[10].

(2) Rainfall pattern redistribution. Let $\mathbf{w}^{(m)} = (w_1^{(m)}, \ldots, w_L^{(m)})$ be a mode-specific nonnegative weight vector (front-loaded, middle-loaded, back-loaded, uniform) satisfying $\sum_{t=1}^{L} w_t^{(m)} = 1$. The redistributed rainfall is

$$P'_t = \Big(\sum_{j=1}^{L} P_j\Big) w_t^{(m)}. \tag{S3}$$

This conserves total rainfall volume while altering its temporal distribution within the input window[11]. We note that this intervention represents an idealized redistribution rather than a physically simulated storm evolution, and is intended to isolate model sensitivity to intra-window timing patterns.

(3) Streamflow-input scaling (state perturbation). Let Q_t^{in} denote the lagged streamflow input channel. With scaling factor β,

$$Q_t^{\text{in}\prime} = \beta Q_t^{\text{in}}, \qquad t = 1, \dots, L, \qquad \beta \in \{0.1, 0.5, 0.9, 1.1, 1.5, 1.9\}. \tag{S4}$$

This isolates sensitivity to hydrological state magnitude in the full lookback window while preserving the original temporal pattern[12].

(4) Temperature increase perturbation. For each temperature-related predictor T_t, we apply

$$T_t' = T_t + \Delta T, \qquad \Delta T \in \{3, 5, 7\}^\circ\text{C}. \tag{S5}$$

This probes thermal forcing effects (e.g., evapotranspiration or snowmelt processes), particularly in energy-limited or snow-influenced catchments[13].

In total, the intervention set contains $4 + 4 + 6 + 3 = 17$ scenarios per station. For each station and scenario, responses are computed against the same baseline prediction, enabling paired comparison[14].

Evaluation metrics

For station i and scenario s, with sample-wise observations $y_{i,n}$ and predictions $\hat{y}_{i,n}^{(s)}$ ($n = 1, \dots, N_i$), we evaluate the following metrics[15, 16]: **(1) MSE.**

(2) Predicted streamflow volume.

$$V_i^{(s)} = \sum_{n=1}^{N_i} \hat{y}_{i,n}^{(s)}. \tag{S6}$$

(3) Predicted peak timing index.

$$\tau_i^{(s)} = \arg \max_{1 \le n \le N_i} \hat{y}_{i,n}^{(s)}. \tag{S7}$$

Peak-timing-based diagnostics are commonly used to characterize flow timing shifts under altered hydro-climatic forcing[17].

Relative to baseline $s = 0$, we define

$$\Delta \text{MSE}_i^{(s)} = \text{MSE}_i^{(s)} - \text{MSE}_i^{(0)}, \quad \Delta V_i^{(s)} = V_i^{(s)} - V_i^{(0)}, \quad \Delta \tau_i^{(s)} = \tau_i^{(s)} - \tau_i^{(0)}. \tag{S8}$$

For visualization comparability across stations, relative changes are reported as

$$\delta \text{MSE}_i^{(s)}(\%) = 100 \times \frac{\Delta \text{MSE}_i^{(s)}}{\max\left(|\text{MSE}_i^{(0)}|, 1\right)}, \qquad \delta V_i^{(s)}(\%) = 100 \times \frac{\Delta V_i^{(s)}}{\max\left(|V_i^{(0)}|, 1\right)}. \tag{S9}$$

Physical consistency criteria

To assess whether model responses align with expected hydrological behavior, we adopt direction-based consistency criteria[7]:

For rainfall scaling and streamflow-input scaling, expected behavior requires directionally consistent changes in streamflow volume, i.e., ΔV increases for scaling factors greater than 1 and decreases for factors less than 1[10].

For rainfall pattern redistribution and temperature perturbation, expected behavior requires that model performance does not exhibit systematic degradation, i.e., no consistent increase in prediction error across scenarios[9].

Finally, for each intervention family, station-level expectedness is aggregated by majority vote over its scenarios. Statistical significance of paired scenario effects is assessed by two-sided Wilcoxon signed-rank tests, with Benjamini–Hochberg correction for multiple comparisons.

1.3 STEH-Derived Interpretability Enhances Random Forest Predictive Performance

To evaluate whether the hydrological structures revealed by STEH can improve a conventional machine-learning benchmark, we designed a paired station-wise comparison between a *baseline* Random Forest (RF) and a *STEH-informed modified* RF[18]. Both models are trained and tested on the same station, temporal split, and target variable, ensuring that any performance differences can be attributed to the STEH-derived design rather than inconsistencies in data or experimental setup[19].

Let station index be $i \in \{1, \ldots, M\}$, predictors $\mathbf{X}_i$, and discharge target $\mathbf{y}_i$. For each station, we train

$$\hat{\mathbf{y}}_i^{\text{base}} = g_{\phi_i}(\mathbf{X}_i), \qquad \hat{\mathbf{y}}_i^{\text{mod}} = g_{\psi_i}(\widetilde{\mathbf{X}}_i), \tag{S10}$$

where g_{ϕ_i} and g_{ψ_i} denote RF predictors with station-specific fitted parameters.

Experiment design

During model training, the hyperparameters of both Random Forest models are optimized using Bayesian optimisation on the validation set[20], ensuring that performance differences are not attributable to discrepancies in tuning strategy. Specifically, the tuned hyperparameters include tree depth ($\in [2, 8]$), the minimum number of samples required for node splitting ($\in [2, 8]$), the minimum number of samples at leaf nodes ($\in [1, 4]$), and the feature sampling ratio for split selection ($\in [0.5, 0.8]$), further information can be seen in Supplementary Notes 2.2.

The task is formulated as a one-step-ahead prediction problem, where a fixed 10-day input window is used to predict next-day discharge. Both baseline and modified models use the same input length ($L = 10$). Data are split chronologically into training, validation, and testing sets with ratios of $0.7/0.15/0.15$, respectively, on a per-station basis. Both models use identical raw drivers, including precipitation, temperature, shortwave radiation, and antecedent discharge.

Baseline RF. The baseline model adopts standard lagged inputs and rolling statistical summaries, without incorporating explicit hydrological regime information.

STEH-informed modified RF. The modified model incorporates structural insights revealed by STEH (primarily P–Q, Q–Q, and T–Q pathways). Specifically, it includes physically interpretable aggregated features (e.g., $P_3, P_7, P_{14}, Q_3, Q_7, Q_3 - Q_7, T_7^+, T_7^-$), introduces regime indicators to distinguish dominant hydrological processes (P–Q, Q–Q, T–Q), and applies light regime-aware sample weighting for selected regimes (P–Q and T–Q) during training.

The dominant regime for each station is identified over the training period based on correlation analysis[11]. Specifically, we compute

$$r_{\text{PQ}} = \text{corr}(P_t, Q_t), \quad r_{\text{QQ}} = \text{corr}(Q_{t-1}, Q_t), \quad r_{\text{TQ}} = \text{corr}(T_t, Q_t), \tag{S11}$$

and define the regime as

$$\mathcal{R}_i = \arg\max\{r_{\text{PQ}}, r_{\text{QQ}}, r_{\text{TQ}}\}. \tag{S12}$$

This regime label is then used as part of the model input.

Training and evaluation protocol

Model performance is evaluated using Nash–Sutcliffe efficiency (NSE), mean squared error (MSE), and Kling–Gupta efficiency (KGE), with test NSE serving as the primary comparison metric[16]. Evaluation is conducted on stations where the sparse-transformer benchmark achieves NSE greater than 0.5[21].

For the selected stations, we report the mean NSE, mean MSE and mean KGE of both baseline and modified RF models, and perform a paired t-test on station-wise NSE differences.

2 Model Hyperparameters

2.1 STEH

Hyperparameters were systematically optimized within the predefined search spaces listed in Table S3[20], ensuring reproducibility and minimizing subjective bias. The sparsity-related settings are consistent with prior sparse-Transformer practice[22].

Due to the large number of study sites, reporting all optimized hyperparameter configurations in this Supplementary Information is impractical. Therefore, the complete set of best hyperparameters is provided in an repository for full transparency and reproducibility:https://github.com/Polyethylenekjc/Sparse-Transformer-In-Streamflow-Prediction/blob/main/outcomes/STEH_Best_Hyperparams.csv

Table S3: Hyperparameter settings and search space

Hyperparameter	**Description**	**Search Space**
Learning rate	Controls optimisation step size	10^{-4} – 10^{-2}
Batch size	Number of samples per mini-batch	32 – 256
Hidden size	Dimension of hidden representations	16 – 256
Number of layers	Number of Transformer encoder layers	1 – 16
Weight sparsity	Sparsity level in weight pruning	0.7 – 0.95
Activation sparsity	Sparsity level for activations	0.6 – 0.9
Gate regularisation	Strength of gate sparsity regularization	10^{-5} – 10^{-3}

2.2 Random Forest

Hyperparameters of the Random Forest models were systematically optimized using Bayesian optimisation within the predefined search spaces listed in Table S4[20], ensuring reproducibility and minimizing subjective bias.

Due to the large number of study sites, reporting all optimized hyperparameter configurations in this Supplementary Information is impractical. Therefore, the complete set of best hyperparameters is provided in a repository for full transparency and reproducibility: https://github.com/Polyethylenekjc/Sparse-Transformer-In-Streamflow-Prediction/blob/main/outcomes/Rf_Best_Hyperparams.csv

Table S4: Hyperparameter settings and search space for Random Forest

Hyperparameter	**Description**	**Search Space**
Tree depth	Maximum tree depth	2 – 8
Min samples (split)	Minimum number of samples required to split	2 – 8
Min samples (leaf)	Minimum number of samples per leaf	1 – 4
Feature ratio	Fraction of features considered per split	0.5 – 0.8

Note: "Baseline" refers to the standard Random Forest model without incorporating STEH-derived hydrological knowledge, while "Modified" denotes the Random Forest model augmented with STEH-informed structured features.

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